

Q1: Explain the difference between coherent and non-coherent detection.

2. M-Array PSK

Q2: Calculate the bandwidth efficiency of an M-ary PSK system with $M=16$.

3. Signal Mapping

Q3: How are signals mapped to orthogonal vectors in the geometric representation of signals?

4. QAM (Quadrature Amplitude Modulation)

Q4: In a QAM system, 16-QAM can transmit how many bits per symbol?

5. AWGN Channel Performance

Q5: Derive the bit error rate (BER) expression for BPSK in an AWGN channel.